

Results and Performance Analysis

This section presents the software and hardware results obtained for the proposed Adaptive Precision Reconfigurable TinyML CNN Accelerator. The performance of the proposed design was evaluated using both TensorFlow-based software experiments and FPGA implementation in Xilinx Vivado. The software evaluation includes the classification accuracy of different quantized models, while the hardware evaluation includes FPGA resource utilization, timing performance, operating frequency, and power consumption. The obtained results are also compared with previously reported CNN accelerator designs to demonstrate the effectiveness of the proposed architecture.

Quantization Results

To reduce the memory requirement and computational complexity of the CNN model, quantization was performed using TensorFlow Lite. Three different precision levels, namely 8-bit, 4-bit, and 1-bit, were evaluated. The objective was to identify a precision level that provides a good balance between model accuracy and hardware efficiency.

The experimental results show that the 8-bit model maintains the original classification accuracy, while the 4-bit model achieves almost the same performance with a much smaller model size. Although the 1-bit model significantly reduces the memory requirement, its classification accuracy decreases considerably. Therefore, the 4-bit model was selected as the best trade-off for hardware implementation.

Table 1. Quantization Results

Precision	Accuracy (%)	Model Size (KB)	Observation
8-bit	100.00	812	Highest accuracy
4-bit	99.94	406	Best balance between accuracy and model size
1-bit	28.25	101	Lowest memory but poor classification accuracy

FPGA Resource Utilization

The proposed CNN accelerator was synthesized and implemented using the Xilinx Vivado Design Suite by targeting the Artix-7 XC7A35T FPGA device. The resource utilization report shows that the proposed design occupies only a very small portion of the available FPGA resources. This indicates that the accelerator has a lightweight architecture and provides sufficient room for implementing additional CNN layers or future enhancements.

Table 2. FPGA Resource Utilization

Resource	Used	Available	Utilization
Slice LUTs	175	20800	0.84%
Slice Registers	68	41600	0.16%
Slices	63	8150	0.77%
LUT as Logic	167	20800	0.80%
LUT as Memory	8	9600	<0.1%
Bonded IOBs	35	210	16.7%
BUFGCTRL	1	32	3.1%

Hierarchy							
Name	Slice LUTs (20800)	Slice Registers (41600)	Slice (8150)	LUT as Logic (20800)	LUT as Memory (9600)	Bonded IOB (210)	BUFGCTRL (32)
▼ cnn_accelerator	175	68	63	167	8	35	1
mac (mac_unit)	91	0	28	91	0	0	0
mem (input_memory)	8	8	2	0	8	0	0
pool (pooling)	73	0	33	73	0	0	0

Figure 4. FPGA Resource Utilization Report generated in Xilinx Vivado.

The MAC unit consumes the largest number of LUTs because multiplication and accumulation are the major computational operations of the CNN accelerator. Even with multiple hardware modules, the overall FPGA utilization remains below 1%, demonstrating the compact nature of the proposed design.

Timing Analysis

Timing analysis was carried out after FPGA implementation to verify whether the proposed design satisfies the specified clock constraint of 100 MHz. During the initial implementation, setup timing violations were observed due to long combinational paths. To overcome this problem, pipeline registers were introduced between different processing stages of the datapath.

After pipelining, all setup, hold, and pulse-width timing requirements were successfully satisfied.

Table 3. Timing Analysis Results

Parameter	Result
Target Clock Frequency	100 MHz
Estimated Maximum Frequency	121.2 MHz
Worst Negative Slack (WNS)	+1.751 ns
Total Negative Slack (TNS)	0 ns
Worst Hold Slack (WHS)	+0.070 ns
Hold Violations	0
Timing Violations	None

   Clock Summary				
Name	Waveform	Period (ns)	Frequency (MHz)	
clk	{0.000 5.000}	10.000	100.000	

Figure 5. Clock Summary showing successful operation at 100 MHz.

The positive setup and hold slack values indicate that the proposed CNN accelerator successfully satisfies all user-defined timing constraints and can reliably operate at the required clock frequency.

Power Analysis

Power analysis was performed using the switching activity obtained from behavioral simulation. A total of 10,000 randomly generated test cases were applied to capture realistic signal transitions. The generated switching activity was used for estimating the on-chip power consumption in Vivado.

The total power consumption of the proposed accelerator is only 79 mW, making it suitable for energy-efficient TinyML and edge AI applications.

Table 4. Power Analysis Results

Parameter	Value
Total On-Chip Power	79 mW
Static Power	60 mW
Dynamic Power	18 mW
Clock Power	1 mW
Signal Power	2 mW
Logic Power	2 mW
I/O Power	14 mW
Junction Temperature	25.4°C
Thermal Margin	74.6°C

Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power: 0.079 W
Design Power Budget: Not Specified
Process: typical
Power Budget Margin: N/A
Junction Temperature: 25.4°C
 Thermal Margin: 74.6°C (15.5 W)
 Ambient Temperature: 25.0 °C
 Effective θ_{JA} : 4.8°C/W
 Power supplied to off-chip devices: 0 W
 Confidence level: Low

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power

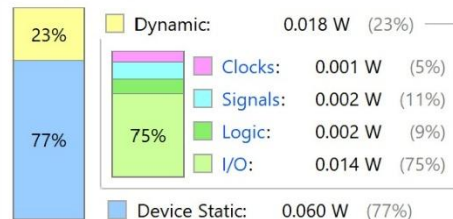


Figure 6. On-chip power analysis report generated in Xilinx Vivado.

The power report shows that static power contributes the largest portion of the total power consumption because the implemented design occupies only a small portion of the FPGA resources. The dynamic power remains very low, confirming that the proposed accelerator is suitable for low-power edge devices.

Module-wise Resource Utilization

The hierarchical utilization report provides information about the FPGA resources consumed by individual modules of the proposed architecture.

Table 5. Module-wise FPGA Resource Utilization

Module	LUTs	Registers
MAC Unit	91	—
Input Memory	8	8
Pooling Unit	73	—
CNN Accelerator (Top Module)	175	68

The MAC unit requires the largest number of LUTs because it performs multiplication and accumulation operations. The input memory uses a small number of memory resources, while the pooling unit requires additional comparison logic. Overall, the complete accelerator occupies only a small portion of the FPGA resources.

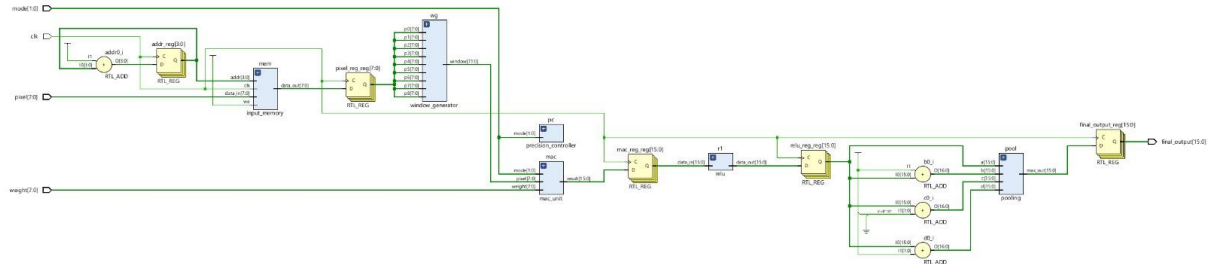


Figure 7. RTL Schematic of the Proposed CNN Accelerator

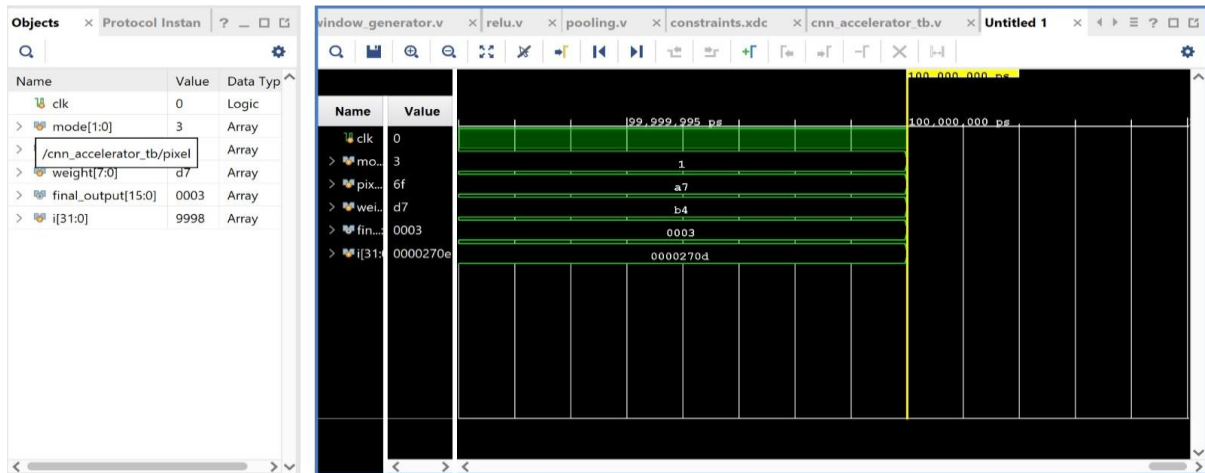


Figure 8. Behavioral Simulation Waveform

Comparison with Existing CNN Accelerator Designs

To evaluate the performance of the proposed Adaptive Precision Reconfigurable TinyML CNN Accelerator, it was compared with recently published FPGA-based CNN accelerator designs. The comparison includes implementation technology, precision level, classification accuracy, PSNR, power consumption, operating frequency, FPGA resource utilization, DSP usage, BRAM utilization, and application domain. Table 5 presents the comparison between the proposed work and existing CNN accelerator architectures reported in the literature.

Table 5. Comparison of Existing CNN Accelerators with Proposed Work

Paper	Year	Technology (FPGA/ASIC)	Precision	Accuracy (%)	PSNR (dB)	Power	Frequency (MHz)	LUTs/Area	DSPs	BRAM	Application
RAMAN: A Re-configurable and Sparse tinyML Accelerator for Inference on Edge	2023	Efinix Ti60 FPGA	2b, 4b, 8b	93% (DS-CNN), ~95% (MobileNetV1)	–	132–137 mW	75	37.2K LUTs	61	118–168	Keyword Spotting, Visual Wake Words
Flare: An FPGA-Based Full Precision Low Power CNN Accelerator with Reconfigurable Structure	2024	XC7A100T / XCZU15EG FPGA	32-bit Floating Point	–	–	1.617–7.329 W	200–300	101.44K Logic Cells	240–3528	–	AlexNet, VGG16
Efficient Dynamic Reconfigurable CNN Accelerator for Edge Intelligence Computing on FPGA (EDRCA)	2023	Xilinx KV260 FPGA	32-bit Floating Point	–	–	5.520 W	250	116,416 LUTs	1248	144	YOLOv2 Tiny Object Detection
Proposed Adaptive Precision Reconfigurable TinyML CNN Accelerator	2026	Xilinx Artix-7 FPGA (XC7A35T)	1-bit, 4-bit, 8-bit	99.94	7.29 (4-bit)	79 mW	100	175 LUTs	0	0	Gesture Recognition (TinyML Edge AI)

Discussion

Table 5 compares the proposed CNN accelerator with recently published FPGA-based CNN accelerators. Most of the existing designs are developed for complex applications such as object detection, image classification, keyword spotting, and visual wake-word detection.

These accelerators mainly use fixed-precision or floating-point computation and are implemented on high-end FPGA devices. As a result, they require a large number of FPGA resources, higher power consumption, and more DSP and BRAM resources. Although these designs provide high computational performance, they are generally intended for large-scale AI applications rather than lightweight TinyML systems.

The proposed Adaptive Precision Reconfigurable TinyML CNN Accelerator is designed specifically for low-power edge AI applications. Unlike conventional fixed-precision architectures, it supports three different precision modes (1-bit, 4-bit, and 8-bit), allowing the precision to be selected according to application requirements. The 4-bit quantized model achieves 99.94% classification accuracy while significantly reducing model size and computational complexity. The FPGA implementation requires only 175 LUTs, 68 registers, consumes only 79 mW of total power, and successfully operates at 100 MHz with no timing violations. These results show that the proposed design achieves an effective balance between accuracy, power consumption, hardware utilization, and operating frequency.

Advantages of the Proposed Work

The proposed accelerator offers several advantages over the existing CNN accelerator designs. First, it provides adaptive precision support, which allows the hardware to operate in 1-bit, 4-bit, or 8-bit modes depending on the application requirements. Second, the hardware architecture is lightweight and occupies less than 1% of the available FPGA resources, making it suitable for resource-constrained embedded systems. Third, the accelerator consumes only 79 mW of total power while maintaining high classification accuracy, making it suitable for battery-powered TinyML devices. Finally, the design successfully meets all timing constraints at 100 MHz, demonstrating reliable real-time operation. These features make the proposed accelerator a practical solution for low-power gesture recognition and future edge AI applications.